

Escape O.B.T.

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Introduction:

[In a few paragraphs totaling about ½ page, introduce the high-level concept of your program. What this looks like depends a lot on what type of thing you are making. An introduction for an *application* will look different than one for a *game*. In general, your introduction should address questions like these:

What does your program do?

What problem does it solve? Why did you write it?

What is the story?

What are the rules? What is the goal?

Who would want to use your program?

What are the primary features of your program?]

On November 9, 1983, The Ticklers invaded Ohio and destroyed all signs of human civilization... except for you. Now they patrol the streets of Ohio, making sure no one escapes. You are currently stuck in Ohio and need to find a way out so that you can live your life in tickle-less bliss.

You move around in a scrolling, grid-based maze, or the streets of Ohio. You have limited sight, because it's dark outside and hard to see, which makes it easier to evade the ticklers. There are multiple mazes in Ohio that you need to navigate through. In each maze, you need to avoid the ticklers while collecting key fragments. Once you find all of the key fragments in the level, find the exit door to progress to the next maze. After every maze, you are closer to escaping Ohio, and the mazes get harder.

Most people these days struggle, sitting in their homes all day, losing their survival instincts. This game heightens their senses and keeps them ready to escape Ohio when needed. This project was created because we wanted to help people improve their survival skills, as people (in movies at least) are really bad at making decisions that impact their survival positively.

People in need of a dopamine boost or just want to have fun would want to use our program.

Instructions:

[Which keyboard keys will do what?

Where will you need to click?

Will you have menus that need to be navigated? What will they look like?

Do actions need to be taken in a certain order?]

[WASD] - Move

[SHIFT] - Sneak: Move around much slower but entities will be less aware of you

[ESC] - Exit to the main menu

Click the PLAY button to enter the first level.

Click the LEVELS button to enter any level you have finished already.

Click the INSTRUCTIONS button for the instructions.

Features List (THE ONLY SECTION THAT CANNOT CHANGE LATER):

Must-have Features:

- The player moves through a scrolling grid-based maze and can move smoothly between tiles in the grid (rather than moving from tile to tile). They can see all of the tiles in an area around their character.
- Key fragments are hidden throughout the maze. The player needs to navigate through the maze and collect all of them. Once they have all the necessary key fragments, they can find the exit tile and move on to the next level.
- Entities cause your character to die immediately upon contact. These entities move slowly and randomly without any regard for the player until they are a few tiles away from them. Then, the entities speed up slightly and move directly towards the player. They cannot be eliminated.
- The player can sneak, which makes them move much slower. While sneaking, entities can't see the player until they are directly next to each other. The player's area of vision decreases while sneaking, and can only sneak for a few seconds before getting up again. There is a small cooldown between each time the player sneaks.
- The levels are sourced from a text file, which is then processed to generate the level. Different characters in the text file relate to different items in-game, such as walls, the player, and enemy entities.

Want-to-have Features:

- The player has three hearts and loses one upon contact with monsters. Once fully depleted, the game ends, and the player has to restart the level. If low on hearts, health kit items are hidden throughout the maze that can be used to regenerate hearts.
- A lantern item is located somewhere near the start of the first level, which allows the player to see a bigger area around themselves. The lantern can be equipped simply by walking over it.
- The player can find weapons to attack the entities and kill them instead of evading or outrunning them. These weapons shoot out projectiles that are aimed in the direction the player is facing, and deal damage to the entities.
- As the player progresses through the levels, the entities get faster. They start off moving about half as fast as the player and eventually go just slightly slower than the player. The entities also get stronger by taking longer to kill.
- The game has background music. As monsters get closer to the player, eerie ambient sounds start playing in the background to alert the player. Sound effects also play for most actions in the game, such as picking up items, shooting the monster, dying, getting hit, walking, and more. While sneaking, sound effects are enhanced.

Stretch Features:

- When entering a level, there is another screen where the player selects the difficulty: easy, normal, or hard. Depending on the level of difficulty, the entities will vary in speed and strength, the player will have different amounts of hearts.
- The player has a strength meter, which increases progressively throughout the game and can be further increased by picking up workout equipment hidden throughout the maze. The strength meter increases the player's agility, their health, and their range of sight.
- There is a separate mode called "INFINITE" where levels are randomly generated. After completing a level, a new level is randomly generated. Once the player dies, the amount of levels they passed is recorded as their score. They have a high score in this mode, which is their goal to beat every time they enter into the mode.

Class List:

- Main - Starts the game
- DrawingSurface - Controls the game's screens and drawing
- Player - User-controlled player that navigates through the maze
- Sprite - Represents every object in the game
- Tickler - Monster that attacks the player
- KeyFragment - Part of a key that needs to be collected to escape each maze
- Wall - A wall that cannot be moved over by anything
- Weapon - Item used to kill ticklers
- Exit - The exit to a level
- Screen - Template for screens
- MenuScreen - Screen shown at the start of the game
- GameScreen - Screen that displays the gameplay
- LevelScreen - Screen of all levels that the player has completed
- DeathScreen - Screen displayed after the player dies
- ScreenSwitcher - Interface that controls switching between the game's screens
- Level - Handles everything having to do with a level

Credits:

- [List the group members and describe how each member contributed to the completion of the final program. This could be classes written, art assets created, leadership/organizational skills exercises, or other tasks. Before you start coding the project, this section should describe *how you plan on splitting the work*. As the project progresses, it should transform into a *description of who completed each task*.
- Give credit to all outside resources used. This includes downloaded images or sounds, external java libraries, parent/tutor/student coding help, etc.]
- Websites - <https://www.artstation.com/artwork/EaJJ6N> (Door image), <https://www.dcode.fr/maze-generator> (Maze generation), <https://www.remove.bg/> (Background remover), <https://www.bing.com/images/create> (AI image creation)
- Nikunj - Javadocs and classes

- Boon - Javadoc, UML, majority of coding
- Yash - UML, audio creation, image creation,